

Bounds, Signs, and the Variable of Integration: Find the Error

Answer Key

AP Calculus AB/BC

Quick Answers

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|--|---|
| 1. -3 | distance on the second interval $= \frac{7}{3}$, (d) total |
| 2. (b) corrected value $= -21$, — | distance $= \frac{23}{3}$, — |
| 3. (b) corrected value at $x = 2 = 13$, — | 6. (b) corrected value $= 0.5$, — |
| 4. (b) corrected value $= 78$, — | 7. (b) corrected area $= 8$, — |
| 5. (b) distance on the first interval $= \frac{16}{3}$, (c) | 8. (b) corrected value $= 2$, — |
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What is verified

10 of 18 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-6 – problems 2, 3, 4, 6, 8

LIM-5 – problems 1, 5, 7

Difficulty 1–5 of 5 across 18 tagged checks.

Common wrong answers

1. If they got 5: added the region below the axis as positive area instead of subtracting it
 2. If they got 21: integrated from the smaller limit to the larger one, flipping the sign
 3. If they got 4: substituted the lower limit into the dummy variable instead of the upper limit into x
 4. If they got 2: kept the original x -limits after substituting $u = x^2 + 1$
 5. If they got -3 : reported the signed integral, which is the displacement
 6. If they got 1: replaced $x \, dx$ by du , dropping the factor of one half
 7. If they got 0: integrated the signed function, so the two halves cancelled
 8. If they got 6: evaluated the u -integral at the old x -limits
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1. Evaluate $\int_0^3 (2x - 4) \, dx$.

Solution. $\int (2x - 4) dx = x^2 - 4x$, so the value is $(9 - 12) - (0 - 0)$.

$$\boxed{-3}$$

The integrand is negative on $[0, 2)$ and positive on $(2, 3]$, and the negative part wins. A student who reports 5 has added the two regions as areas instead of letting the signs cancel.

2. $\int_5^2 2x dx$; the student answered 21. (a) Name the error. (b) Corrected value.

Solution (a) — instructor-judged. The limits were read in the wrong order. The integral runs *from 5 to 2*, and $\int_b^a = -\int_a^b$. Credit any wording that names the reversed limits or states the answer must be the negative of the student's.

Solution (b). $[x^2]_5^2 = 4 - 25$.

$$\boxed{-21}$$

3. $\frac{d}{dx} \int_1^x (3t^2 + 1) dt$; the student answered $3t^2 + 1$ and then 4. (a) Name the error. (b) Corrected value at $x = 2$.

Solution (a) — instructor-judged. The variable t is a dummy: it exists only inside the integral and cannot appear in the answer. The Fundamental Theorem substitutes the *upper limit* into the integrand, giving $3x^2 + 1$. The student instead substituted the lower limit, $t = 1$. Credit any wording that identifies t as a dummy variable.

Solution (b). $3(2)^2 + 1 = 12 + 1$.

$$\boxed{13}$$

4. $\int_0^2 x(x^2 + 1)^3 dx$ by $u = x^2 + 1$; the student answered 2. (a) Name the error. (b) Corrected value.

Solution (a) — instructor-judged. The limits were never converted. When x runs from 0 to 2, $u = x^2 + 1$ runs from 1 to 5; the student integrated in u but evaluated at the old x -limits. Converting the limits, or converting back to x before evaluating, both fix it.

Solution (b).

$$\frac{1}{2} \int_1^5 u^3 du = \frac{1}{8} [u^4]_1^5 = \frac{625 - 1}{8}.$$

$$\boxed{78}$$

5. $v(t) = t^2 - 4$ m/s on $[0, 3]$; the student gave -3 metres as the total distance. (a) The error. (b) $\int_0^2 (4 - t^2) dt$. (c) $\int_2^3 (t^2 - 4) dt$. (d) Total distance.

Solution (a) — instructor-judged. The signed integral of velocity is *displacement*, not distance travelled. The velocity changes sign at $t = 2$, so the trip has to be split there and the two pieces added as positive amounts. Credit any wording that distinguishes displacement from total distance.

Solution (b). On $[0, 2]$ the particle moves backwards, so the distance is the integral of $-v$: $\left[4t - \frac{t^3}{3}\right]_0^2 = 8 - \frac{8}{3}$.

$$\boxed{\frac{16}{3}}$$

Solution (c). On $[2, 3]$ it moves forwards: $\left[\frac{t^3}{3} - 4t\right]_2^3 = (9 - 12) - \left(\frac{8}{3} - 8\right)$.

$$\boxed{\frac{7}{3}}$$

Solution (d). Adding the two positive pieces:

$$\boxed{\frac{23}{3}}$$

About 7.67 metres — and note the displacement is still -3 , which is the number the student reported.

6. $\int_0^{\sqrt{\pi/2}} x \cos(x^2) dx$; the student replaced $x dx$ by du and got 1. (a) Name the error. (b) Corrected value.

Solution (a) — instructor-judged. With $u = x^2$ we have $du = 2x dx$, so $x dx = \frac{1}{2} du$. Replacing $x dx$ by du throws away a factor of one half and doubles the answer. Credit any wording that names the missing one half.

Solution (b). With the limits converted (u runs from 0 to $\frac{\pi}{2}$),

$$\frac{1}{2} \int_0^{\pi/2} \cos u du = \frac{1}{2} [\sin u]_0^{\pi/2}.$$

$$\boxed{\frac{1}{2}}$$

7. Area between $y = x^3$ and the x -axis on $[-2, 2]$; the student answered 0. (a) The error. (b) Corrected area.

Solution (a) — instructor-judged. The signed integral cancels the piece below the axis against the piece above it, which is why it comes out zero — but area is never negative. Split at $x = 0$ and add the magnitudes. Credit any wording that separates the signed integral from geometric area.

Solution (b). By symmetry the two pieces are equal:

$$\int_{-2}^2 |x^3| dx = 2 \int_0^2 x^3 dx = 2 \left[\frac{x^4}{4} \right]_0^2 = 2(4).$$

$$\boxed{8}$$

8. $\int_1^4 x^{-1/2} dx$ by $u = \sqrt{x}$; the student answered 6. (a) Name the error. (b) Corrected value. (c) The habit that prevents it.

Solution (a) — instructor-judged. When x runs from 1 to 4, $u = \sqrt{x}$ runs from 1 to 2. The student rewrote the integrand in u but left the x -limits in place, so the integral was only half converted.

Solution (b). Either finish in u , $\int_1^2 2 \, du = 2(2 - 1)$, or return to x : $[2\sqrt{x}]_1^4 = 4 - 2$. Both give

$$\boxed{2}$$

Solution (c) — instructor-judged. Convert the limits at the same moment you convert the integrand, so the integral is never left half in one variable and half in the other; equivalently, undo the substitution and evaluate in the original variable. Full credit needs a stated rule, not a restatement of the mistake.